



Orthogonality

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Orthogonality



Orthogonal Sets

Definition

- A set of vectors $\{a_1, \dots, a_k\}$ in R^n is **orthogonal** set if each pair of distinct vectors is orthogonal (**mutually orthogonal vectors**).

A basis B of an inner product space V is called an **orthonormal basis** of V if

- a) $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ for all $\mathbf{v} \neq \mathbf{w} \in B$, and (mutual orthogonality)
- b) $\|\mathbf{v}\| = 1$ for all $\mathbf{v} \in B$. (normalization)

- ❑ set of n -vectors $a_1, \dots, a_k$ are (*mutually*) *orthogonal* if $a_i \perp a_j$ for $i \neq j$
- ❑ They are *normalized* if $\|a_i\| = 1$ for $i = 1, \dots, k$
- ❑ They are *orthonormal* if both hold
- ❑ Can be expressed using inner products as

$$a_i^T a_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Orthogonal Sets

- Geometry
- Algebra



<https://youtu.be/dqdSzqsm7bY>

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal** (to each other) if $\mathbf{u} \cdot \mathbf{v} = 0$.

Suppose V is an inner product space.

Two vectors $\mathbf{v}, \mathbf{w} \in V$ are called **orthogonal** if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$.

The Pythagorean Theorem

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$

Orthogonal Sets

Example

- ❑ Zero vector is orthogonal to every vector in vector space V
- ❑ The standard basis of $\mathbb{R}^n$ or $\mathbb{C}^n$ is an orthogonal set with respect to the standard inner product.

Orthogonal Sets

Theorem

If $S = \{a_1, \dots, a_k\}$ is an orthogonal set of nonzero vectors in R^n , then S is linearly independent and is a basis for the subspace spanned by S .

Proof

If $k = n$, then prove that S is a basis for R^n

Linear combinations of orthonormal vectors

Corollary

□ A simple way to check if an n -vector y is a linear combination of the orthonormal vectors $a_1, \dots, a_k$, if and only if:

$$y = (a_1^T y)a_1 + \dots + (a_k^T y)a_k$$

□ For orthogonal vectors $a_1, \dots, a_k$:

$$y = c_1 a_1 + \dots + c_k a_k$$

$$c_j = \frac{y \cdot a_j}{a_j \cdot a_j}$$

Orthonormal vectors

Theorem

If $S = \{a_1, \dots, a_k\}$ is an orthogonal set of nonzero vectors in R^n , then S is linearly independent and is a basis for the subspace spanned by S .

Proof

If $k = n$, then prove that S is a basis for R^n

Orthonormal vectors

Theorem

Independence-dimension inequality

If the n -vectors $a_1, \dots, a_k$ are linearly independent, then $k \leq n$.

- Orthonormal sets of vectors are linearly independent
- By independence-dimension inequality, must have $k \leq n$
- When $k = n$, $a_1, \dots, a_n$ are an *orthonormal basis*

Example

❑ Standard unit n-vectors $e_1, \dots, e_n$

❑ The 3-vectors

$$\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

❑ The 2-vectors shown below



❑ The standard basis in $P_n(x) [-1,1]$ (be the set of real-valued polynomials of degree at most n.)

Linear combinations of orthonormal vectors

Example

Write x as a linear combination of a_1, a_2, a_3 ?

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad a_1 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \quad a_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad a_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

02

Orthogonal Subspaces



Definition

- Two subspaces W_1 and W_2 of the same space V are orthogonal, denoted by $W_1 \perp W_2$, if and only if each vector $w_1 \in W_1$ is orthogonal to each vector $w_2 \in W_2$ for all w_1, w_2 in W_1, W_2 respectively:
- $$\langle w_1, w_2 \rangle = 0$$

Example

If the bases of two subspaces are orthogonal, it implies that the subspaces themselves are orthogonal.

03

Orthogonal Complements



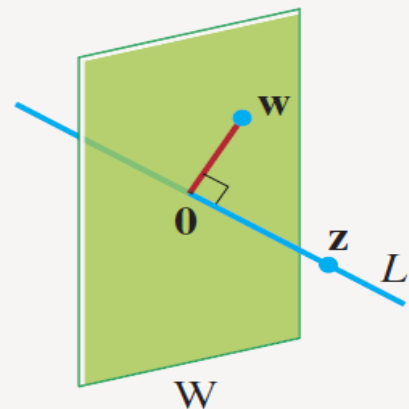
Definition

- If a vector z is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then z is said to be orthogonal to W .
- **The set of all vectors z that are orthogonal to W is called the orthogonal complement of W and is denoted by $W^\perp$**

Example

W be a plane through the origin in $\mathbb{R}^3$.

$$L = W^\perp \text{ and } W = L^\perp$$



Orthogonal Complements

Theorem

$W^\perp$ is a subspace of $\mathbb{R}^n$.

Theorem

$W^\perp \cap W = \{\mathbf{0}\}$.

Important

We emphasize that W_1 and W_2 can be orthogonal without being complements.
 $W_1 = \text{span}((1, 0, 0))$ and $W_2 = \text{span}((0, 1, 0))$.

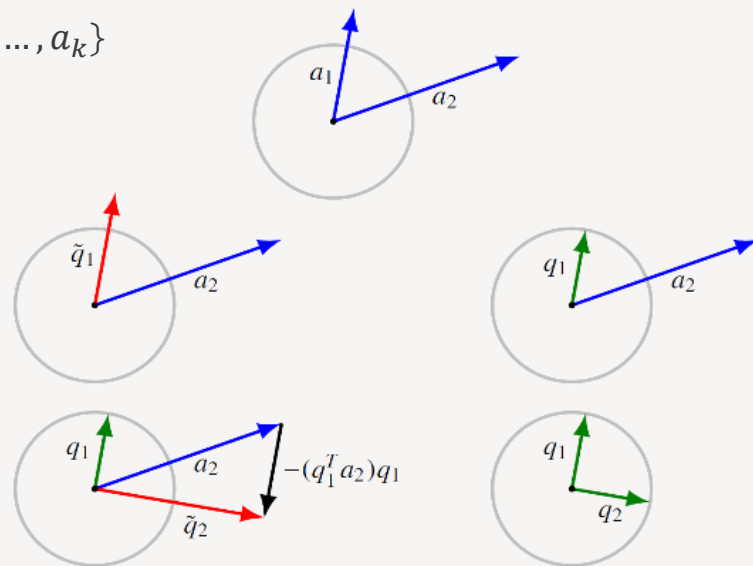
04

Gram–Schmidt Algorithm



Gram–Schmidt (orthogonalization) algorithm

- Find orthonormal basis for $\text{span} \{a_1, a_2, \dots, a_k\}$
- Geometry:



Gram–Schmidt (orthogonalization) algorithm

- Find orthonormal basis for $\text{span} \{a_1, a_2, \dots, a_k\}$
- Algebra:


$$1) q_1 = \frac{a_1}{\|a_1\|}$$

$$2) \widetilde{q}_2 = a_2 - (q_1^T a_2)q_1 \rightarrow q_2 = \frac{\widetilde{q}_2}{\|\widetilde{q}_2\|}$$

$$3) \widetilde{q}_3 = a_3 - (q_1^T a_3)q_1 - (q_2^T a_3)q_2 \rightarrow q_3 = \frac{\widetilde{q}_3}{\|\widetilde{q}_3\|}$$

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$$k) \widetilde{q}_k = a_k - (q_1^T a_k)q_1 - \dots - (q_{k-1}^T a_k)q_{k-1} \rightarrow q_k = \frac{\widetilde{q}_k}{\|\widetilde{q}_k\|}$$

Gram–Schmidt (orthogonalization) algorithm

Example

Find orthogonal set for $a = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $c = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

Gram–Schmidt (orthogonalization) algorithm

□ Why $\{q_1, q_2, \dots, q_k\}$ is a orthonormal basis for $\text{span}\{a_1, a_2, \dots, a_k\}$?

- $\{q_1, q_2, \dots, q_k\}$ are normalized.
- $\{q_1, q_2, \dots, q_k\}$ is a orthogonal set
- a_i is a linear combination of $\{q_1, q_2, \dots, q_i\}$

$$\text{span}\{q_1, q_2, \dots, q_k\} = \text{span}\{a_1, a_2, \dots, a_k\}$$

□ q_i is a linear combination of $\{a_1, a_2, \dots, a_i\}$

Gram–Schmidt (orthogonalization) algorithm

□ Given n -vectors $a_1, \dots, a_k$ for $i = 1, \dots, k$

1. Orthogonalization: $\tilde{q}_i = a_i - (q_1^T a_i)q_1 - \dots - (q_{i-1}^T a_i)q_{i-1}$
2. Test for linear dependence: if $\tilde{q}_i = 0$, quit
3. Normalization: $q_i = \frac{\tilde{q}_i}{\|\tilde{q}_i\|}$

Note

- If G–S does not stop early (in step 2), $a_1, \dots, a_k$ are linearly independent.
- If G–S stops early in iteration $i = j$, then a_j is a linear combination of $a_1, \dots, a_{j-1}$ (so $a_1, \dots, a_k$ are linearly dependent)

$$a_j = (q_1^T a_j)q_1 + \dots + (q_{j-1}^T a_j)q_{j-1}$$

Complexity of Gram–Schmidt algorithm

- Gram-Schmidt algorithm gives us an explicit method for determining if a list of vectors is linearly dependent or independent.
- What is complexity and number of flops for this algorithm?
 - $O(nk^2)$ why?
- Given n -vectors $a_1, \dots, a_k$ for $i = 1, \dots, k$

1. Orthogonalization: $\tilde{q}_i = a_i - (q_1^T a_i)q_1 - \dots - (q_{i-1}^T a_i)q_{i-1}$
2. Test for linear dependence: if $\tilde{q}_i = 0$, quit
3. Normalization: $q_i = \frac{\tilde{q}_i}{\|\tilde{q}_i\|}$

Complexity of the Gram–Schmidt algorithm. We now derive an operation count for the Gram–Schmidt algorithm. In the first step of iteration i of the algorithm, $i - 1$ inner products

$$q_1^T a_i, \dots, q_{i-1}^T a_i$$

between vectors of length n are computed. This takes $(i - 1)(2n - 1)$ flops. We then use these inner products as the coefficients in $i - 1$ scalar multiplications with the vectors $q_1, \dots, q_{i-1}$. This requires $n(i - 1)$ flops. We then subtract the $i - 1$ resulting vectors from a_i , which requires another $n(i - 1)$ flops. The total flop count for step 1 is

$$(i - 1)(2n - 1) + n(i - 1) + n(i - 1) = (4n - 1)(i - 1)$$

flops. In step 3 we compute the norm of $\tilde{q}_i$, which takes approximately $2n$ flops. We then divide $\tilde{q}_i$ by its norm, which requires n scalar divisions. So the total flop count for the i th iteration is $(4n - 1)(i - 1) + 3n$ flops.

The total flop count for all k iterations of the algorithm is obtained by summing our counts for $i = 1, \dots, k$:

$$\sum_{i=1}^k ((4n - 1)(i - 1) + 3n) = (4n - 1) \frac{k(k - 1)}{2} + 3nk \approx 2nk^2,$$

where we use the fact that

$$\sum_{i=1}^k (i - 1) = 1 + 2 + \dots + (k - 2) + (k - 1) = \frac{k(k - 1)}{2}, \quad (5.7)$$

which we justify below. The complexity of the Gram–Schmidt algorithm is $2nk^2$; its order is nk^2 . We can guess that its running time grows linearly with the lengths

Orthonormal basis

Corollary

Every finite-dimensional inner product space has an orthonormal basis.



Conclusion

Existence of Orthonormal Bases

- ❑ Every finite-dimensional inner product space has an orthonormal basis.
- ❑ Since finite-dimensional inner product spaces (by definition) have a basis consisting of finitely many vectors, and the Gram–Schmidt process tells us how to convert that basis into an orthonormal basis, we now know that every finite-dimensional inner product space has an orthonormal basis.



Example

Find an orthonormal basis for $P_2(x)$ in $[-1, 1]$ with respect to the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx$$



05

Projection



Projection

- Finding the distance from a point B to line l = Finding the length of line segment BP
- AP : projection of AB onto the line l



If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$ and $\mathbf{u} \neq \mathbf{0}$, then the **projection of $\mathbf{v}$ onto $\mathbf{u}$** is the vector $proj_{\mathbf{u}}(\mathbf{v})$ defined by

$$proj_{\mathbf{u}}(\mathbf{v}) = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{u} \cdot \mathbf{u}} \right) \mathbf{u}$$



The projection of $\mathbf{v}$ onto $\mathbf{u}$

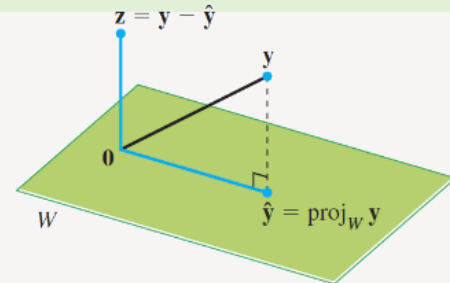
Orthogonal Decomposition Theorem

Theorem

Let W be a subspace of V . Then for any vector y in V , there exists a **unique vector $\hat{y}$ in W** , and a **unique vector z in $W^\perp$** , such that $y = \hat{y} + z$. The vector w is called the orthogonal projection of v onto W .

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \cdots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p$$

Proof



The orthogonal projection of y onto W .

06

Orthogonal Matrix

Square Orthogonal Matrix

Note

- ❑ Columns of A are orthonormal $\leftrightarrow A^T A = I$
- ❑ Square matrix with orthonormal columns is an orthogonal matrix
 - Columns and rows are orthonormal vectors
 - $A^T A = A A^T = I$
 - Is necessarily invertible with inverse $A^T = A^{-1}$

Square Orthogonal Matrix

Example

□ Identity matrix $I^T I = I$

□ Rotation matrix

$$R^T R = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} =$$

$$\begin{bmatrix} \cos^2 \theta + \sin^2 \theta & -\cos \theta \sin \theta + \sin \theta \cos \theta \\ -\sin \theta \cos \theta + \cos \theta \sin \theta & \sin^2 \theta + \cos^2 \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Square Orthogonal Matrix

Example

□ Reflection matrix

$$\begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}^T \begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix} =$$

$$\begin{bmatrix} \cos^2(2\theta) + \sin^2(2\theta) & \cos(2\theta)\sin(2\theta) - \sin(2\theta)\cos(2\theta) \\ \sin(2\theta)\cos(2\theta) - \cos(2\theta)\sin(2\theta) & \sin^2(2\theta) + \cos^2(2\theta) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Theorem

Every 2×2 orthogonal matrices can be expressed as Rotation or Reflection, or a composition of them.

proof?

Tall Orthogonal Matrix

- $A \in \mathbb{R}^{m \times n}, m > n$
- The inner products of the columns give the identity, so $A^T A = I_n$

$$(A^T A)_{ij} = \langle a_i, a_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

- AA^T is a **projection matrix** onto the column space of A . If $m > n$ then AA^T is not full rank. Therefore, $AA^T \neq I_m$

$$\text{rank}(AA^T) \leq \min(\text{rank}(A), \text{rank}(A^T)) = \text{rank}(A) \leq n$$

But remember: $AA^T \in \mathbb{R}^{m \times m}$. So the maximum possible rank of AA^T is n , which is strictly less than m .

Wide Orthogonal Matrix

- $A \in \mathbb{R}^{m \times n}, m < n$
- The inner products of the rows give the identity, so $AA^T = I_m$

$$(AA^T)_{ij} = \langle a_i, a_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

- $A^T A$ is a **projection matrix** onto the row space of A . If $m < n$ then $A^T A$ is not full rank. Therefore, $A^T A \neq I_n$

$$\text{rank}(A^T A) \leq \min(\text{rank}(A), \text{rank}(A^T)) = \text{rank}(A) \leq m$$

But remember: $A^T A \in \mathbb{R}^{n \times n}$.

So the maximum possible rank of $A^T A$ is m , which is strictly less than n .

Properties of Orthogonal Matrix

Note

If $A \in \mathbb{R}^{m \times n}$ has orthonormal columns, then the linear function $f(x)=Ax$:

- ❑ Preserves inner product:

$$(Ax)^T(Ay) = x^T y$$

- ❑ Preserves norm:

$$\|Ax\| = \|x\|$$

- ❑ Preserves distances:

$$\|Ax - Ay\| = \|x - y\|$$

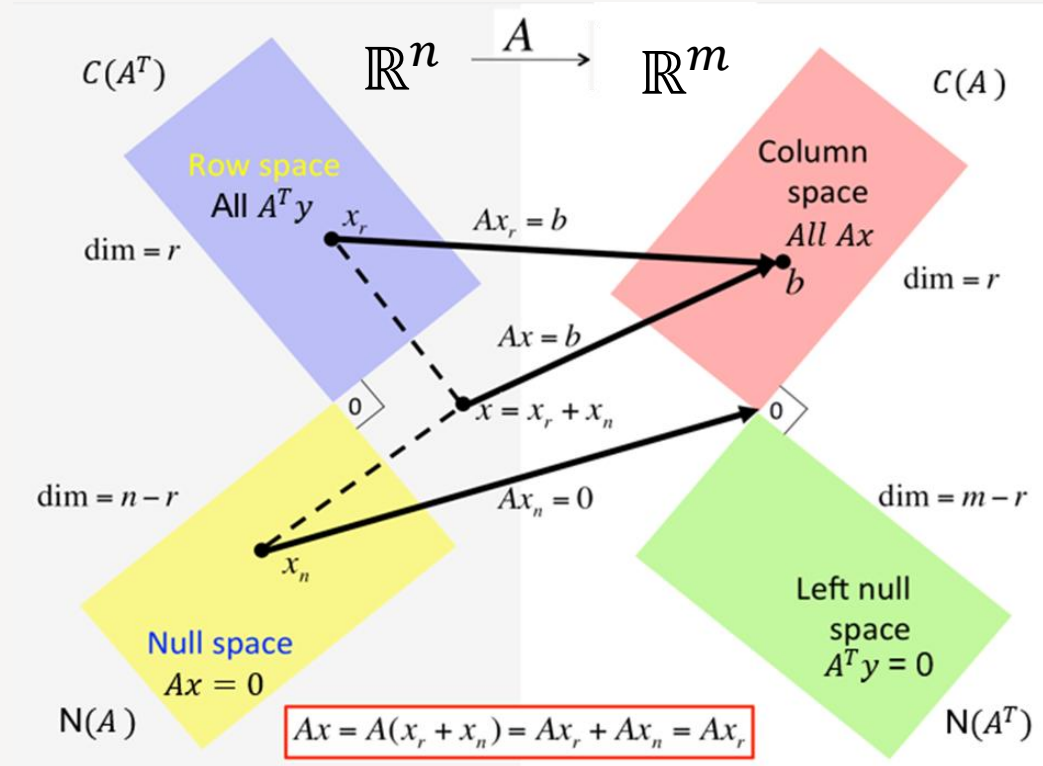
- ❑ Preserves angles:

$$\angle(Ax, Ay) = \arccos\left(\frac{(Ax)^T(Ay)}{\|Ax\|\|Ay\|}\right) = \arccos\left(\frac{x^T y}{\|x\|\|y\|}\right) = \angle(x, y)$$

This is a mapping with preserving properties of input

Four Fundamental Subspaces of Matrix Space

- **Proof that:**
 - $C(A^T) \perp N(A)$
 - $C(A) \perp N(A^T)$



07

QR Factorization (QR Decomposition)

Gram–Schmidt in Matrix Notation

Important

Run Gram-Schmidt on columns $a_1, \dots, a_k$ of $n \times k$ matrix A :

$$\tilde{q}_1 = a_1, \quad q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|}$$
$$\Rightarrow a_1 = \|\tilde{q}_1\|q_1$$

$$\tilde{q}_2 = a_2 - (q_1^T a_2)q_1, \quad q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|}$$
$$\Rightarrow a_2 = (q_1^T a_2)q_1 + \|\tilde{q}_2\|q_2$$

$\vdots$

$$\tilde{q}_i = a_i - (q_1^T a_i)q_1 - \dots - (q_{i-1}^T a_i)q_{i-1}, \quad q_i = \frac{\tilde{q}_i}{\|\tilde{q}_i\|}$$
$$a_i = (q_1^T a_i)q_1 + \dots + (q_{i-1}^T a_i)q_{i-1} + \|\tilde{q}_i\|q_i$$

Review

- Matrix-Matrix Multiplication

As a set of matrix-vector products.

$$C = AB = A \begin{bmatrix} | & | & \cdots & | \\ b_1 & b_2 & \cdots & b_p \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ Ab_1 & Ab_2 & \cdots & Ab_p \\ | & | & \cdots & | \end{bmatrix}$$

Here the i th column of C is given by the matrix-vector product with the vector on the right, $c_i = Ab_i$. These matrix-vector products can in turn be interpreted using both viewpoints given in the previous subsection.

- Matrix-Vector Multiplication

If we write A by columns, then we have:

$$y = Ax = \begin{bmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = [a_1]x_1 + [a_2]x_2 + \cdots + [a_n]x_n .$$

- y is a linear combination of the columns A .

Gram–Schmidt in Matrix Notation

Important

$$a_1 = \|\tilde{q}_1\|q_1$$

$$a_2 = (q_1^T a_2)q_1 + \|\tilde{q}_2\|q_2$$

$$\vdots$$

$$a_k = (q_1^T a_k)q_1 + \cdots + (q_{k-1}^T a_k)q_{k-1} + \|\tilde{q}_k\|q_k$$

$$[a_1 \quad a_2 \quad \cdots \quad a_k] = [q_1 \quad q_2 \quad \cdots \quad q_k] \begin{bmatrix} \|\tilde{q}_1\| & q_1^T a_2 & \cdots & q_1^T a_k \\ 0 & \|\tilde{q}_2\| & \cdots & q_2^T a_k \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & q_{k-1}^T a_k \\ 0 & 0 & \cdots & \|\tilde{q}_k\| \end{bmatrix}$$

$$A_{n \times k} = Q_{n \times k} \times R_{k \times k}$$

Gram–Schmidt in Matrix Notation

Important

1. Run Gram-Schmidt on columns $a_1, \dots, a_k$ of $n \times k$ matrix A
2. If columns are linearly independent, get orthonormal $q_1, \dots, q_k$
3. Define $n \times k$ matrix Q with columns $q_1, \dots, q_k$
4. $Q^T Q = I$
5. From Gram-Schmidt algorithm

$$\begin{aligned} a_i &= (q_1^T a_i)q_1 + \dots + (q_{i-1}^T a_i)q_{i-1} + \|\tilde{q}_i\|q_i \\ &= R_{1i}q_1 + \dots + R_{ii}q_i \end{aligned}$$

With $R_{1j} = q_1^T a_j$ for $i < j$ and $R_{ii} = \|\tilde{q}_i\|$

6. Defining $R_{ij} = 0$ for $i > j$ we have $A = QR$
7. R is upper triangular, with positive diagonal entries

QR Factorization

Definition

A factorization of a matrix A as $A = QR$ where Factors satisfy $Q^T Q = I$, R upper triangular with positive diagonal entries, is called a QR factorization of A .

Suppose A is a square matrix with linearly independent columns. Then there exist unique matrices Q and R such that Q is unitary, R is upper triangular with only positive numbers on its diagonal, and

$$R_{jk} = \langle a_k, q_j \rangle$$

$$A = QR.$$

Note

The QR factorization of a matrix :

- ❑ Can be computed using Gram-Schmidt algorithm (or some variations)
- ❑ Has a huge number of uses, which we'll see soon

QR Factorization

Important

To find QR decomposition:

□ Q : Use Gram-Schmidt to find orthonormal basis for column space of A

□ Let $R = Q^T A$

□ OR: $R_{jk} = \langle a_k, q_j \rangle$

□ If A is a square matrix, then Q is square with orthonormal columns (orthogonal matrix)

QR Factorization

Theorem

if $A \in \mathbb{R}^{m \times n}$ has linearly independent columns then it can be factored as

$$A = QR$$

Q-factor

- Q is $m \times n$ with orthonormal columns ($Q^T Q = I$)
- If A is square ($m = n$), then Q is orthogonal ($Q^T Q = Q Q^T = I$)

R-factor

- R is $n \times n$, upper triangular, with nonzero diagonal elements
- R is nonsingular (diagonal elements are nonzero)

QR Decomposition

Example

$$A = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix}$$

$$q_1 = \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}, q_2 = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, q_3 = \frac{1}{2} \begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}, \|\tilde{q}_1\| = 2, \|\tilde{q}_2\| = 2, \|\tilde{q}_3\| = 4$$

□ QR :

$$\begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 2 & 8 \\ 0 & 0 & 4 \end{bmatrix}$$

Generalization of QR Factorization

$$A_{4 \times 6} = [\underline{a_1} \quad \underline{a_2} \quad a_3 \quad \underline{a_4} \quad a_5 \quad a_6]$$

Linear Independent

$$\begin{cases} a_1 = a_{11}q_1 \\ a_2 = a_{21}q_1 + a_{22}q_2 \\ a_3 = a_{31}q_1 + a_{32}q_2 \\ a_4 = a_{41}q_1 + a_{42}q_2 + a_{43}q_3 \\ a_5 = a_{51}q_1 + a_{52}q_2 + a_{53}q_3 \\ a_6 = a_{61}q_1 + a_{62}q_2 + a_{63}q_3 \end{cases}$$

Block upper triangular matrix

$$[a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \quad a_6] = [q_1 \quad q_2 \quad q_3] \begin{bmatrix} a_{11} & a_{21} & a_{31} & a_{41} & a_{51} & a_{61} \\ 0 & a_{22} & a_{32} & a_{42} & a_{52} & a_{62} \\ 0 & 0 & 0 & a_{43} & a_{53} & a_{63} \end{bmatrix}$$

$$A_{4 \times 6} = Q_{4 \times 3} \times R_{3 \times 6}$$

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